11.24

a. Since p-value<0.05, MIXED is significant at the 5% level.

F-value>10

```
            Estimate Std. Error t value  Pr(>|t|)
(Intercept) -0.359568   0.079154 -4.5426 1.502e-05 ***
mon         -0.108248   0.103389 -1.0470   0.29753
tue         -0.066092   0.101035 -0.6542   0.51446
wed         -0.049274   0.103774 -0.4748   0.63591
thu          0.039347   0.100705  0.3907   0.69681
stormy       0.446339   0.079211  5.6348 1.508e-07 ***
mixed        0.236875   0.078513  3.0170   0.00321 **
```

```
> cat("F-stat:", f_test_a$F[2], "\n")
F-stat: 16.14003
```

b. The standard errors of each variables decrease.

```
(b) Demand 2SLS (IV = STORMY + MIXED):
> print(coeftest(iv_b))

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  8.540234   0.156608 54.5325 < 2.2e-16 ***
lprice      -0.930141   0.353399 -2.6320  0.009768 **
mon         -0.011902   0.208371 -0.0571  0.954559
tue         -0.525832   0.202295 -2.5993  0.010684 *
wed         -0.562620   0.206958 -2.7185  0.007674 **
thu          0.099871   0.202845  0.4923  0.623501
```

```
> print(coeftest(iv_b_table11_5))

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  8.505911   0.166167 51.1890 < 2.2e-16 ***
lprice      -1.119417   0.428645 -2.6115  0.010333 *
mon         -0.025402   0.214774 -0.1183  0.906077
tue         -0.530769   0.208000 -2.5518  0.012157 *
wed         -0.566351   0.212755 -2.6620  0.008989 **
thu          0.109267   0.208787  0.5233  0.601837
```

c. Apply Sargan test. $NR^2 = 0.772$, p-value=0.380

=>Non-reject H_0, Surplus IV is valid.

```
(c) Sargan test (manual):
> cat("N*R^2 =", sargan_stat, "\n")
N*R^2 = 0.7721695
> cat("p-value (chi^2 df=1):", p_sargan, "\n")
p-value (chi^2 df=1): 0.3795467
```

d. The day-of-week dummies (MON-THU) are jointly insignificant in the reduced-form equation for ln(PRICE) (F = 0.628, p = 0.644). This implies they do not directly affect price through the supply side, so they can serve as valid IVs for ln(PRICE) in the supply equation. We can now estimate the supply equation by 2SLS with confidence.

```
> cat("F-stat:", f_test_d$F[2], "  p-value:", f_test_d$`Pr(>F)`[2], "\n")
F-stat: 0.6278532   p-value: 0.6436925
```

11.28

a.

Demand: P = a1 + a2*Q + a3*PS + a4*DI + e (a2 < 0, a3 > 0, a4 > 0)

Supply: P = b1 + b2*Q + b3*PF + e (b2 > 0, b3 > 0)

b. The signs are correct. They are all significantly different from zero at 5% level.

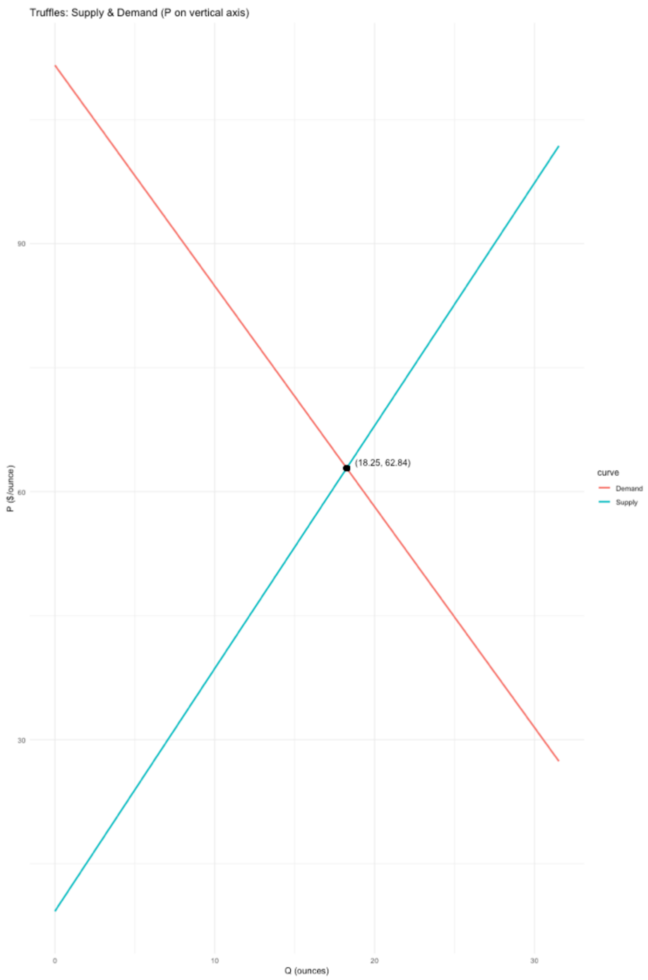
```
(b) Demand 2SLS  (P on LHS):
> print(coeftest(dem_2sls))

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept) -11.4284    13.5916 -0.8408  0.408103
q            -2.6705     1.1750 -2.2729  0.031535 *
ps            3.4611     1.1156  3.1025  0.004582 **
di           13.3899     2.7467  4.8749 4.675e-05 ***
```

```
Supply 2SLS  (P on LHS):
> print(coeftest(sup_2sls))

t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -58.79822    5.85916 -10.035 1.316e-10 ***
q             2.93671    0.21577  13.610 1.321e-13 ***
pf            2.95849    0.15596  18.969 < 2.2e-16 ***
```

c. elasticity = -1.272464

```
(c) Price elasticity of demand at means:
> cat("a2' =", a2_prime, "  mean(P) =", mean_p, "  mean(Q) =", mean_q, "\n")
a2' = -2.670519   mean(P) = 62.724   mean(Q) = 18.45833
> cat("elasticity =", elas_d, "\n")
elasticity = -1.272464
```

d.



e. The structural 2SLS equilibrium (Q = 18.250, P = 62.843) is virtually identical to the reduced-form prediction (Q = 18.260, P = 62.815). This agreement confirms the mathematical equivalence between 2SLS and Indirect Least Squares (ILS) in a just-identified system — solving the structural equations for equilibrium is algebraically the same as plugging the exogenous values into the reduced-form predictions.

```
(e) Reduced-form prediction at DI*=3.5, PS*=22, PF*=23:
> cat(sprintf("  Structural (2SLS) : P = %.3f, Q = %.3f\n", P_eq, Q_eq))
  Structural (2SLS) : P = 62.843, Q = 18.250
> cat(sprintf("  Reduced-form pred : P = %.3f, Q = %.3f\n", P_rf, Q_rf))
  Reduced-form pred : P = 62.815, Q = 18.260
```

f. OLS yields a wrong-signed and insignificant coefficient on Q in the demand equation (+0.151, p = 0.764), a classic symptom of simultaneity bias since P and Q are jointly determined. The supply OLS estimate (+2.661) is closer to the 2SLS result (+2.937) but still biased downward. Overall, the demand equation suffers far more severe endogeneity than the supply equation, justifying the use of IV estimation.

```
> cat("\n--- Coefficient comparison (q) ---\n")

--- Coefficient comparison (q) ---
> cat(sprintf("Demand  2SLS q: %8.4f    OLS q: %8.4f\n",
+             coef(dem_2sls)["q"], coef(dem_ols)["q"]))
Demand  2SLS q:  -2.6705    OLS q:   0.1512
> cat(sprintf("Supply  2SLS q: %8.4f    OLS q: %8.4f\n",
+             coef(sup_2sls)["q"], coef(sup_ols)["q"]))
Supply  2SLS q:   2.9367    OLS q:   2.6613
```